

RDBMS Using ORACLE-2022

Q1-A

- 1.DBMS-Database Management System
- 2.TCL-Transaction Control Language
- 3.DML-Data Manipulation Language
- 4.DQL-Data Query Language

Q1-B

1. Formatting command SQL “COMPUTE SUM”

- The "COMPUTE SUM" clause in SQL is not a standard SQL command, but it might be specific to certain database systems or query tools.
- It's used for calculating the sum of values in a result set and displaying the total as a separate row in the query result.
- EX:-

```
SELECT product_id, sales_amount
FROM sales
COMPUTE SUM(sales_amount);
```

2. Formatting command SQL “BREAK ON”

- The "BREAK ON" command is not a standard SQL command, but rather a command that is specific to certain database systems, particularly Oracle SQL*Plus.
- It is used for formatting the output of SQL queries by controlling when breaks or summary lines should be displayed in the query result.
- EX:-

```
BREAK ON department_id SKIP 1
SELECT department_id, employee_id, salary
FROM employees
ORDER BY department_id, employee_id;
```

Q1-C

1. Characteristics of RDBMS

- RELATIONSHIP BETWEEN TWO TABLES OR FILE CAN BE SPECIFIED AT THE TIME OF TABLE CREATION.
- SUPPORTS CLIENT/SERVER ARCHITECTURE
- SUPPORTS DISTRIBUTED DATABASES
- MULTIPLE SECURITY LEVES; 1 LOGIN AT OS LEVEL 2 COMMAND LEVEL SECURITY 3 OBJECT LEVEL SECURITY
- MANY TABLES ARE GROUPED IN ONE DATABASE IN RDBMS

2. BCNF

- BCNF, or Boyce-Codd Normal Form, is a higher level of database normalization that is based on the functional dependencies of attributes within a relational database. BCNF is an important concept in database design and ensures that a database is structured in such a way that it avoids certain types of anomalies and redundancies.
- **Functional Dependency:** In a relational database, an attribute A is said to be functionally dependent on attribute B if, for every value of B, there is only one corresponding value of A. In other words, if you know the value of B, you can determine the value of A.
- BCNF, we need to satisfy the following two conditions:
- **The table must be in 1NF (First Normal Form):** This means that the table must have a primary key, and each attribute in the table must be atomic (indivisible).
- **For every non-trivial functional dependency $X \rightarrow Y$, X must be a superkey:** In BCNF, a non-trivial functional dependency means that Y is not a subset of X, and X is not a superkey. A superkey is a set of attributes that can be used to uniquely identify a tuple in a table.

Q1-D

1.Note On SQL *PLUS

- SQL stands for Structured Query Language.
- It is a domain-specific language used for managing and manipulating relational databases.
- SQL allows you to interact with databases to perform various tasks, including creating and modifying database structures, inserting, updating, and deleting data, and retrieving data for analysis and reporting.
- **Key Components of SQL:**
- SQL comprises several key components:
- **Data Definition Language (DDL):** DDL statements are used to define and manage the structure of a database. Common DDL statements include CREATE TABLE, ALTER TABLE, and DROP TABLE.
- **Data Manipulation Language (DML):** DML statements are used to interact with and manipulate data within a database. Common DML statements include SELECT, INSERT, UPDATE, and DELETE.

- **Data Query Language (DQL):** DQL is a subset of DML that specifically deals with querying data. The most common DQL statement is SELECT, which retrieves data from one or more tables.
- **Data Control Language (DCL):** DCL statements are used to control access and permissions within a database. Common DCL statements include GRANT and REVOKE.
- **Basic SQL Commands:**
 - SELECT: Used to retrieve data from one or more tables. It is often the most frequently used SQL command and supports various clauses such as WHERE, ORDER BY, and GROUP BY.
 - INSERT: Adds new rows of data to a table.
 - UPDATE: Modifies existing data in a table.
 - DELETE: Removes data from a table.
 - CREATE TABLE: Creates a new table with specified columns and data types.
 - ALTER TABLE: Modifies the structure of an existing table, such as adding, modifying, or deleting columns.
 - DROP TABLE: Deletes an entire table and its data.
- **Data Integrity:**
 - SQL databases enforce data integrity through various constraints, including:
 - Primary Key: Ensures that each row in a table is unique and has a unique identifier.
 - Foreign Key: Maintains referential integrity between tables by ensuring that values in one table correspond to values in another.
 - Unique Constraint: Enforces uniqueness of values within a column or a combination of columns.
 - Check Constraint: Allows you to define custom rules that data must adhere to.
- **SQL Database Management Systems (DBMS):**
 - SQL is used with various database management systems, such as MySQL, PostgreSQL, Oracle, Microsoft SQL Server, and SQLite. Each DBMS may have its own dialect of SQL, although many SQL statements are standardized.
- **Advanced SQL Topics:**
 - Joins: Combining data from multiple tables using JOIN operations (e.g., INNER JOIN, LEFT JOIN).
 - Subqueries: Embedding one SQL query within another.
 - Views: Virtual tables created from SQL queries, simplifying complex queries or providing a level of abstraction.

- Stored Procedures and Functions: Reusable code blocks written in SQL for data manipulation and automation.
- Transactions: A sequence of one or more SQL statements that are executed as a single unit of work, ensuring data consistency.

2. Dr. E.F. CODD RULES

- Rule 0 – Foundation rule
 - Any relational database management system that is propounded to be RDBMS or advocated to be a RDBMS should be able to manage the stored data in its entirety through its relational capabilities.
- Rule 1 - Rule of Information
 - Relational Databases should store the data in the form of relations. Tables are relations in Relational Database Management Systems.
- Rule 2 – Rule of Guaranteed Access
 - The use of pointers to access data logically is strictly forbidden. Every data entity which is atomic in nature should be accessed logically by using a right combination of the name of table, primary key represented by a specific row value and column name represented by attribute value.
- Rule 3 – Rule of Systematic Null Value Support
 - Null values are completely supported in relational databases. They should be uniformly considered as 'missing information'. Null values are independent of any data type.
- Rule 4 – Rule of Active and online relational Catalogue
 - In the Database Management Systems lexicon, 'metadata' is the data about the database or the data about the data. The active online catalogue that stores the metadata is called 'Data dictionary'.
- Rule 5 – Rule of Comprehensive Data Sub-language
 - A single robust language should be able to define integrity constraints, views, data manipulations, transactions and authorizations
- Rule 6 – Rule of Updating Views
 - Views should reflect the updates of their respective base tables and vice versa. A view is a logical table which shows restricted data. Views generally make the data readable but not modifiable. Views help in data abstraction.
- Rule 7 – Rule of Set level insertion, update and deletion
 - A single operation should be sufficient to retrieve, insert, update and delete the data.
- Rule 8 – Rule of Physical Data Independence

- Batch and end user operations are logically separated from physical storage and respective access methods.
- Rule 9 – Rule of Logical Data Independence
- Batch and end users can change the database schema without having to recreate it or recreate the applications built upon it.
- Rule 10 – Rule of Integrity Independence
- Integrity constraints should be available and stored as metadata in data dictionary and not in the application programs.
- Rule 11 – Rule of Distribution Independence
- The Data Manipulation Language of the relational system should not be concerned about the physical data storage and no alterations should be required if the physical data is centralized or distributed.
- Rule 12 – Rule of Non Subversion
- Any row should obey the security and integrity constraints imposed. No special privileges are applicable

Q2-A

- 1. Query within Query known as_____.
- Subquery
- 2. _____ command is used to insert data into the table.
- insert into tablename values('.....');
- 3. Join is command the data from two or more tables.(True/False).
- True
- 4. DELETE table command cannot be rollback(True/False).
- False

Q2-B

1. Explain COMMIT and ROLLBACK.

- COMMIT is a transaction control language in SQL. It lets a user permanently save all the changes made in the transaction of a database or table. Once you execute the COMMIT, the database cannot go back to its previous state in any case.
- When transaction begins its assigned to a rollback segment. Rollback segments store the “before” image of data which is ready to run in oracle .
- When creating image of data some time more memory spend to data which is useless so rollback segment undo these extra memory space and keep data in their perfect size.
- Oracle 9i allows you to undo to be managed in either of two ways:
- Manually
- Automatically

2. Explain GRANT and REVOKE commands.

- GRANT & REVOKE are the popular members of SQL family. these are the type of DCL commands that are used to assign permission to the users to perform a different task. The GRANT command is used for permitting the users whereas the REVOKE command is used for removing the authorization.
- When you create any object like table or etc you are owner of that object. So if you want other user use your object or they want to use your object you need to give them permission to access this is called granting of Privileges .
- Syntax of Granting permission using Grant statement :-
- Grant <object privileges>
- On <object name>
- To <UserName>
- [With Grant Option]
- Privileges once given can be taken back by the owner of the object This is call Revoking of Privileges.
- Syntax of Revoking permission :
- Revoke <object privileges>
- On <object Name>
- From <UserName>

Q2-C

1. Explain Having and Group by clause.

- The SQL HAVING Clause
- Having clause special use for give condition to group by like WHERE.
- You use the HAVING clause to filter groups of rows
- GROUP BY can be used without HAVING.
- But HAVING must be used with GROUP BY.
- You place the HAVING clause after your GROUP BY clause:
- HAVING Syntax
- SELECT column_name
FROM table_name
GROUP BY column_name
HAVING condition
ORDER BY column_name;
- The SQL GROUP BY Statement
- The GROUP BY statement groups rows that have the same values into summary rows, like "find the maxsal from each job in emp table".

- The GROUP BY statement is often used with aggregate functions (COUNT, MAX, MIN, SUM, AVG) to group the result-set by one or more columns.
- By default, GROUP BY sorts the rows in ascending order based on the values in the group column.
- GROUP BY Syntax
- SELECT column_name
FROM table_name
GROUP BY column_name
ORDER BY column_name;

2. Explain any three Aggregate Function.

- AVG : returns average value from table
- Select avg(sal) from emp
- MIN : returns minimum value from table
- Select min(sal) from emp
- MAX : returns maximum value from table
- Select max(sal) from emp
- SUM : return sum value
- Select sum(sal) from emp

Q2-D

1. Explain Data constraints in details.

- Constraints are set of rules used to enforce data integrity.
- They don't take space in database like data.
- Even if single column fails to satisfy constrain, entire record will be rejected by Oracle.
- **Types of constraints are :-**
 - Primary key
 - Foreign key
 - Unique
 - NOT NULL
 - Check
- **Feature of primary key:**
 - Does not allow duplicate values
 - NULL values not allowed
 - Primary key is not compulsory but recommended
 - Primary key can not be LONG or LONG RAW data type
 - Only one key is allowed per table
 - Unique Index created automatically on Primary key

- **---primary key at column level----**
- create table uni1
(ccode number(3) primary key,
corsename varchar2(15),
duration number(1));
- **---primary key at table level---**
- create table uni
(ccode number(3),
corsename varchar2(15),
duration number(1),
primary key(ccode,corsename));
- **Feature of foreign key:**
- It references a column of another table and it can be same table also
- Referenced field must be Unique or Primary key. The default is primary key
- Defined foreign key table is called child table where as defined primary key table is called parent table
- Child table may have duplicates and NULLS in foreign key column
- Parent record can be deleted only if child record does not exist
- Parent table can not be updated if child record exist
- **ON DELETE CASCADE :-**
- When the record is deleted in Parent table, all records in Child Table will be deleted automatically.
- **ON DELETE SET NULL :-**
- When record is deleted in Parent table, all field value held by foreign key of child table is set to NULL.
- Entire record will not be deleted.

- **Define foreign key:**
- create table college
(rno number(3),
sname char(20),
course_no number(3) references uni1(ccode));
on delete cascade
on delete set null

2. What is Join ? Explain different type of joins.

- A JOIN clause is used to combine rows from two or more tables, based on a related column between them.

- **Different Types of SQL JOINS**

- Inner Join

- Outer Join

- Left outer Join

- Right outer Join

- Full outer Join

- Self Join

- Cross Join

- **Inner join:**

- The INNER JOIN keyword selects records that have matching values in both tables.

- This join is also known as equi join

- **Outer join:**

- SQL LEFT JOIN Keyword

- The LEFT JOIN keyword returns all records from the left table (table1), and the matched records from the right table (table2).

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- **Full outer join:**

- SQL FULL OUTER JOIN Keyword

- The FULL OUTER JOIN keyword return all records when there is a match in either left (table1) or right (table2) table records.

- **SQL Self JOIN:**

- A self JOIN is a regular join, but the table is joined with itself.

- **Cross join**

- In this join there is no condition given so records multiply by each other table

- Like if one table have 3 records and other table have 4 and we apply cross join than result will be 12 rows display

Q3-A

1. An Insert operation is not allowed when a view is created from multi table. (true/false) - True

2. List the types of snapshot.

Ans. Simple snapshot

Complex snapshot

3. The Oracle Engine automatically locks table data while executing SQL statements is called_____ locking.

Ans.DML

4. A NOWAIT option can be used to inform the oracle engine to terminate the sql statements if the record has already been locked. (true/false).

Ans.True

Q3-B

1. Describe DDL Lock.

- DDL locks (dictionary locks)
- DDL locks protect the structure of schema objects .
- It simply deals with descriptions of the database schema and is used to create and modify the structure of database objects in the database.
- DDL is a set of SQL commands used to create, modify, and delete database structures but not data.

2. Describe Latches.

- Protect internal database structures such as data files. Latches are visible only to the local instance in memory as opposed to locks which maintain information within the oracle database and are visible to all of the instances that have access to the database. As such, locks operate at the database level versus latches which are based at the instance level in oracle memory.

Q3-C

1. Explain lock Escalation.

- Lock escalation is a process that databases use to reduce the number of locks they hold.
- It occurs when a database raises the locks to a higher level on table.
- For example, if a single user locks many rows in a table, some databases automatically escalate the user's row locks to a single table.
- Locks in SQL Server can be quite expensive. For each lock the Lock Manager acquires, SQL Server must reserve memory – 64 bytes or 128 bytes. The amount depends on whether we are dealing with a 32-bit or 64-bit system, respectively.
- As the number of row locks on a table increases, SQL Server must acquire more and more memory. Hence, other processes are starving, out of memory.
- The table lock implies that no other session can modify its rows for the transaction duration, which may happen even when a blocking session does not manipulate all rows in the table.
- It is also worth mentioning that other factors can affect how locks are acquired and escalated in SQL Server. Those are the isolation level configured, indexing, and trace flags.

2. Explain Snapshot.

- Snapshot is a recent copy of the table from the database or a subset of rows/columns of a table.
- The SQL statement that creates and subsequently maintains a snapshot normally reads data from the database residing server.
- A snapshot is created on the destination system with the create snapshot SQL command. The remote table is immediately defined and populated from the master table.
- These are used to dynamically replicate data between distributed databases.
- **Two types of snapshots are available.**
Simple snapshots
Complex snapshots
- **Advantages :**
 - Response time is improved when local read-only copy of table exists.
 - Ease network loads.
 - Data subsetting.
 - Disconnected computing.
 - Mass deployment.
- **Disadvantages:**
 - Snapshots are not reachable when primary database goes offline.
 - It does not support full text indexing.
 - Snapshot runs out of disk if data changes frequently faster.
 - As no.of snapshots increases, disk space becomes problematic.

Q3-D

1. Write a note on bitmap index and application domain index.

- A bitmap is a mapping from one system such as integers to bits.
- It is also known as bitmap index or a bit array.
- The memory is divided into units for bitmap.
- These units may range from a few bytes to several kilobytes.
- Each memory unit is associated with a bit in the bitmap.
- If the unit is occupied, the bit is 1 and if it is empty, the bit is zero.
- The bitmap provides a relatively easy way to keep track of memory as the size of the bitmap is only dependent on the size of the memory and the size of the units.
- **Application Domain Index :**
 - Application domain indexes are custom indexes that are specific to the needs of your application.
 - They can be based on the semantics of your data and queries.

- **Example:**
- Suppose you have a table books with a publication_date column, and you want to create an index that takes into account the availability of books in a specific library branch.
- You could create an application domain index like this:
- CREATE INDEX idx_library_availability ON books (branch_id, publication_date);

Bitmap Indices Example

record number	ID	gender	income_level
0	76766	m	L1
1	22222	f	L2
2	12121	f	L1
3	15151	m	L4
4	58583	f	L3

Bitmaps for gender		Bitmaps for income_level	
m	10010	L1	10100
f	01101	L2	01000
		L3	00001
		L4	00010
		L5	00000

2. Write a note on pessimistic and optimistic locking.

- **Pessimistic**
- A Pessimistic approach is an approach of concurrency control algorithms in which the transaction is delayed if there is a conflict with each other at some point of time in the future.
- It locks the database's record for update access and other users can only access record as read-only or have to wait for a record to be 'unlocked'.
- Programming an app with a pessimistic concurrency approach can be more complicated and complex in managing because of deadlocks' risk.
- **Optimistic**
- An Optimistic approach is an approach of concurrency control algorithms that are based on assumption that conflicts of operations on a database are rare.
- It is advisable to run these transactions to completion and to check for conflicts only before they commit also here there's no checking to be done while the execution of transactions.
- This approach does not need any locking or time-stamping method. In an optimistic approach, a transaction is executed without any problems of restriction until transaction is committed.
- The optimistic approach allows the transactions to proceed in an unsynchronized way and also allows conflict checking at the end.
- This approach is also known as **validation** or **certification approach**.

Q4-A

- 1. Procedural/ structured query language
- 2. False

- 3. National Character Large Object
- 4. - Implicit cursor
- - Explicit cursor

Q4-B

1. Describe use of database trigger.

A database trigger is a predefined set of actions that automatically execute in response to specific events or changes in a database.

- These triggers are typically used to enforce data integrity, automate tasks, or log changes.
- **Syntax :-**
- CREATE OR REPLACE TRIGGER trigger_name
{BEFORE | AFTER | INSTEAD OF} {INSERT | UPDATE | DELETE}
ON table_name
[FOR EACH ROW]
DECLARE
 -- Declaration section
BEGIN
 -- Trigger code
EXCEPTION
 -- Exception handling, if needed
END;

2. Describe for loop in pl/SQL .

- In PL/SQL, a FOR loop is a control structure used to iterate over a range of values or through elements in a collection.
- **Example:-**
- BEGIN
 FOR i IN 1..5 LOOP
 DBMS_OUTPUT.PUT_LINE('Value of i: ' || i);
 END LOOP;
END;
- If you want the output in reverse order you can just add a keyword reverse before specifying the range.

Q4-C

1. Explain function.

- A function is a subprogram that is used to return a single value.
- You must declare and define a function before invoking it.

It can be declared and defined at a same time or can be declared first and defined later in the same block.

- **Syntax:**

- CREATE [OR REPLACE] FUNCTION function_name (parameters)
RETURN return_datatype IS/AS
[declaration_section]
BEGIN
[executable_section]
EXCEPTION
[exception_section]
END [function_name];

- We can also define some parameters while creating function.
- These parameters are:
 1. IN: It is a default parameter. It passes the value to the subprogram.
 2. OUT: It returns a value to the caller.
 3. IN OUT: It passes an initial value to the subprogram and returns an updated value to the caller.

- **Example:**

- create or replace procedure abc(a number, b number) as
add number(5);
function xyz(n1 number,n2 number) return number as
jadd number (10);
begin
jadd:=n1+n2;
return jadd;
end xyz;
begin
add:=xyz(a,b); --Function call
dbms_output.put_line('Add: ' || add);
end;

2. Explain exception handling.

- PL/SQL supports programmers to catch such conditions using EXCEPTION block in the program and an appropriate action is taken against the error condition.
- **There are two types of exceptions :**
 - 1 System-defined exceptions
 - 2 User-defined exceptions

1. system defined:-

No_data_found :-

- In This exception we need not declare exception, we can use directly in our exception part when in our pl/sql block no data match
- This exception generally use for SELECT statement, when no record match we can use this in our exception part

2. user defined:-

- To create user defined exception just follow this steps
- Declacre exception in Declaration part
Example : not_eligible exception;
- Raise Declared exception in Execution part
Example : raise not_eligible;
- In last our Exception part just put your declared exception in between WHEN and THEN for proper message to our exception
Example : when not_eligible then

Q4-D**1. Write a note on objects.**

- In Oracle, "objects" typically refer to database objects.
- They are the entities used to store, manage, and manipulate data within an Oracle database.
- Some common types of database objects in Oracle include:
 1. Table
 2. Views
 3. Sequence
 4. Indexes
 5. Synonyms
- **1. Tables:**
 - Tables are used to store data in rows and columns.
 - They are the fundamental structure for organizing and storing data in a relational database.
 - Example:


```
create table stud
( no number(2),
  name char(20));
```
- **2. Views:**
 - An Oracle view is a virtual table that physically doesn't exist.

- It stores only definition in oracle data dictionary in the form of SQL statement
- Creating view does not take any storage space.
- A view is created by a query joining one or more tables.

- **Example:**

```
create view v1
as select ename, sal,deptno
from emp
where deptno=20;
```

- **3. Sequences:**

- Sequence produces a series of numerical values in accordance with the specification used to create it.
- It automatically generates the primary key and unique key values.
- It may be in ascending or descending order and can be used in multiple tables.

- **Example :**

- create sequence seq1;

- **4.Indexes:**

- Indexes are data structures that improve the retrieval speed of rows from a table.
- They are used to speed up SELECT statements by providing fast access to data based on certain columns.

- **5.Synonyms:**

- Synonyms are aliases for database objects.
- They simplify access to objects and provide a level of abstraction.

2. Write a note on nested table.

- Nested tables are similar to PL/SQL tables, but they can
- be stored in the database as a column in a table or as
- a database object.
- Nested tables are single-dimensional, unbounded collections of homogeneous elements.
- First, a nested table is single-dimensional, meaning that each row has a single column of data like a one-dimension array.
- Second, a nested table is unbounded. It means that the number of elements of a nested table is predetermined.
- Third, homogeneous elements mean that all elements of a nested table have the same data type.

- **example to declare a nested table:-**

- CREATE TABLE Orders
(order_id NUMBER PRIMARY KEY,
products NestedTableType)
NESTED TABLE products STORE AS ProductsTable;

- **example:-**

- -- creation of object:
- create type obj as object
(m1 number(3),m2 number(3),m3 number(3),m4 number(3));
- -- creation of nested table of obj type:
- create type tnest as table of obj;
- -- creation of nested table:
- create table stud_tab
(sname char(30),
smark tnest)
nested table smark store as tnest_tab;
- --Inserting in nested table:
- insert into stud_tab values ('&sname',tnest(obj(&m1,&m2,&m3,&m4)));

Q5-A

- 1. - Backup and Recovery
 - - Space Management
 - - Patching and Upgrades
 - - Security Updates
- 2. True
- 3. SMON- System Monitor Process
- 4. CKPT- Checkpoint process

Q5-B

1. Describe oracle export utility.

- Oracle provides an export utility called "exp" (or "expdp" for Data Pump) allows you to extract data, schemas, tables, and other database objects into a binary export file.
- You can use it to generate a dump file containing data and metadata.
- There are two main versions of the Oracle Export Utility:
 1. Traditional Export (exp)
 2. Data Pump Export (expdp)

2. Describe control files .

- The control files of database is the binary file that contains a great deal of database information.
- The control files contain name , data about database log file .
- Oracle allows you to maintain duplicate copies with the help of control files.

Q5-C

1. Explain table space.

- A table space is a storage structure containing tables, indexes, large objects, and long data.
- They are used to organize data in a database into logical storage groupings that relate to where data is stored on a system.
- Table spaces are stored in database partition groups.
- Table space is logical storage unit within as oracle database.
- Its logical Because a tablespace is not visible in the file system of the machine on which The database stores
- **Create tablespace :**
- Create tablespace <tbasp_name> ' <datafile_path>' Size<n M/K/G>
EXTENT ANAGEMENT LOCAL AUTOLOCATTE;
- **Alter tablespace :**
- ALTER TABLESPACE TABS1 ADD DATAFILE 'PATH' SIZE <N M/K/G>;
- **Drop tablespace :**
- DROP TABLESPACE TABSP1 INCLUDING CONTENTS;

2. Explain initialization parameters

- The initialization parameter file is a text file that contains a list of parameters and a value for each parameter.
- The file should be written in the client's default character set. Specify values in the parameter file which reflect your installation.
- The following are sample entries in a parameter file:
PROCESSES = 100
OPEN_LINKS = 12
GLOBAL_NAMES = TRUE

- **Database administrators can use initialization parameters to do the following:**
- optimize performance by adjusting memory structures, for example, the number of database buffers in memory
- set some database-wide defaults, for example, how much space is initially allocated for a context area when it is created
- set database limits, for example, the maximum number of database users
- specify names of files
- **The following rules govern the specification of parameters in the parameter file:**
- All parameters are optional.
- Only parameters and comments should appear in the parameter file.
- A pound sign (#) starts a comment; the rest of the line is ignored.
- The server has a default value for each parameter. This value may be operating system dependent, depending on the parameter.
- Parameters can be specified in any order.
- Case (upper or lower) in filenames is only significant if case is significant on the host operating system.

Q5-D

1. Write a note on rollback segment.

- In the ORACLE database, the rollback segment is a structure that stores previous data during transactions.
- To support the rollback facility we have a rollback segment in the ORACLE database.
- If during a transaction any failures occur then it Makes sure that changes are not reflected back Into our original Database.
- It helps us in restoring our database in Consistent state in case of any failure. To ensure Data consistency and isolation property.
- The rollback segment is used to recover our data In case of database failure.
- With the help of the Rollback command, we can Restore our previous data which is stored inside The rollback segment to our original database.
- When transaction begins, its assigned to a rollback segment. Rollback segments store the “before” image of data which is Ready to run in oracle.
- When creating image of data some time more memory spend to Data which is useless so rollback segment undo these extra Memory space and keep data in their perfect size.

- **Oracle 9i and above version allows you to undo space to be managed in either of two ways :-**
 1. Manually
 2. Automatically
- **Manually :-**
 - This mode is commonly referred to as rollback managed undo (RMU).
 - Its invoke by setting the parameter UNDO_MANAGEMENT to MANUAL
- **Automatically :-**
 - This mode is commonly referred to as automatically managed undo (AMU). Its invoke by setting the parameter UNDO_MANAGEMENT to AUTO.
- **Create rollback segment :-**
 - CREATE [PUBLIC] ROLLBACK SEGMENT rbs_name option(s)

2. Write a note on schedule architecture.

- Oracle Database includes Oracle Scheduler, an enterprise job scheduler to help you simplify the scheduling of hundreds or even thousands of tasks.
- Oracle Scheduler (the Scheduler) is implemented by the procedures and functions in the DBMS_SCHEDULER PL/SQL package.
- **Job:** A job is a specific task or unit of work that you want to automate using Oracle Scheduler.
- **Program:** A program defines what a job does.. A job references a program to execute a specific task.
- **Schedule:** A schedule defines when and how often a job should run. It specifies the frequency, start time, end time, and recurrence pattern for executing a job.
- **Job Class:** Job classes are used to categorize and manage jobs. You can assign jobs to specific job classes based on their characteristics or priorities.
- **Window:** A window is a specific time interval during which jobs are allowed to run.
- **Dependency:** Dependency defines the relationship between jobs. You can specify that one job should only run after another job has completed successfully.
- "Scheduling Jobs with Oracle Scheduler" involves using Oracle Scheduler, a feature within the Oracle Database, to automate and manage the execution of various tasks or jobs at specified times or under specific conditions.

- **Creating Jobs:** Jobs can include tasks like running SQL scripts, PL/SQL procedures, or external scripts. These jobs are encapsulated as objects within the database.
- **Defining Programs:** For each job, you need to specify a program that defines what the job does. Programs can be created within the database or reference external executables.
- **Setting Schedules:** Once you have defined jobs and programs, you need to create schedules that determine when and how often these jobs should run.
- **Monitoring and Notifications:** Oracle Scheduler provides tools for monitoring the execution of scheduled jobs. You can view job status, logs, and performance metrics.
- **Job History:** Oracle Scheduler maintains a history of job executions, allowing administrators to track the progress of jobs, view job status, and diagnose problems.
- **Security and Access Control:** Access control mechanisms ensure that only authorized users can create, modify, or execute scheduled jobs.